

JPE Replication Report

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1 Paper Identification

ID	20210685
Author	Jin
Title	Buying Data from Consumers: The Impact of Monitoring in U.S. Auto Insurance
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. There are a few glitches in execution, which we are able to fix for you. Please take those onboard and change accordingly.
2. One figure in appendix B looks different for us.
3. The data availability is insufficient in your readme. Please see my separate email about this issue. In a nutshell, you will have to document in title footnote to manuscript and in the readme that the identity of the data provider cannot be revealed, but that I was able to verify both your NDA with them, as well as their actual existence.

3 General

- ☑ Data Availability and Provenance Statements
 - ☑ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☑ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☑ Details on each Data Source
- ☑ Dataset list
- ☑ Computational requirements
 - ☑ Software Requirements
 - ☑ Controlled Randomness (as necessary)
 - ☑ Memory, Runtime, and Storage Requirements
- ☑ Description of programs/code
 - ☑ License for Code (Optional, but recommended)

- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. – As per the [JPE policy](#), "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

{REQUIRED} Please provide a codebook for the data. – The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

- ☐ Dataset is provided in public (dataverse) deposit
- ☒ Dataset is PRIVATELY provided (not for publication)
- ☐ DOI or URL is provided, and works: (PROVIDE URL OR DOI HERE)
- ☒ Access conditions are described:
 - The data is not publicly available and requires a data use agreement to access.

- ☐ Data are not cited, but a working paper, article, or other document is cited > (not a data citation)
- ☐ Data are cited

☐ in the manuscript

☐ in the README

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☐ Authors (with affiliations) are listed in the same order as on the paper:

Note: Affiliations are missing from the README.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:

R (version 4.3.0 or above)

CmdStan (version 2.34 or above)
- ☐ Computational Requirements specified as follows:

Cluster size, etc.
- ☒ Time Requirements specified as follows:

~10–30 hrs

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 3 - Variables flagged in data: 25 - Code files with PII references: 52 - PII references in code: 1739

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	LICENSE.txt	1	son
Data	tab_5.csv	12	block, loc
Data	tab_c2_block2.csv	12	block, loc
Code	bootstrap_helpers.R	25	name, lat

File Type	File	Variables/References	PII Categories
Code	<code>c1_get_rf_exhibits.R</code>	95	lon, son, name, second, city, coord, lat, block, loc, zip
Code	<code>c2_get_param_tables.R</code>	177	lat, name, lon, second, lname, block, loc, zip
Code	<code>c3_get_fitctf_exhibits.R</code>	32	son, lat, lon, name, block, loc, coord
Code	<code>config.R</code>	17	lat, block, loc, son
Code	<code>ctf_calibration.R</code>	10	lat, name, lname
Code	<code>ctf_clear_cache.R</code>	1	name
Code	<code>ctf_equi_k_find_equilibria.R</code>	1	lat
Code	<code>ctf_equi_k_save_grid.R</code>	33	lat, name, block, loc, lname, lon
Code	<code>ctf_preload.R</code>	22	block, loc, lat, lname, name
Code	<code>ctf_table_helpers.R</code>	18	lat, name, lon
Code	<code>extract_model_estimates_cmdstan.R</code>	41	name, block, loc, lat
Code	<code>find_cycles.R</code>	12	lat, name, lon
Code	<code>getX.R</code>	9	name, loc, zip, lat, lname
Code	<code>get_c_latentparams.R</code>	10	lname, name, lat
Code	<code>get_ctf_util_profit.R</code>	57	lat, second, zip, lon
Code	<code>get_d_latentparams.R</code>	7	lat, block, loc
Code	<code>get_d_pred.R</code>	39	lat, block, loc, zip, second
Code	<code>get_fit_cp.R</code>	14	lat, name, lname
Code	<code>get_next_rrev.R</code>	7	lat, name
Code	<code>load_estimation_bootstrap_data.R</code>	22	lat, son, zip, lname, name
Code	<code>load_model_data.R</code>	21	name, block, loc, zip
Code	<code>load_sim_data.R</code>	29	name, lname, loc, zip, son
Code	<code>packages.tex</code>	7	lat, lon, url, son

File Type	File	Variables/References	PII Categories
Code	<code>panel_renw_clean.R</code>	33	zip, sex, school, second, loc, location, lname, name, lat, email, social, city, community
Code	<code>paper.tex</code>	2	lat
Code	<code>paper_setting.tex</code>	6	name, lat, url
Code	<code>run_all.R</code>	13	son, name, lat
Code	<code>run_simulated.R</code>	57	lat, son, name
Code	<code>s0_thanks.tex</code>	3	school, name, son
Code	<code>s1_introduction.tex</code>	7	son, lon, lat
Code	<code>s2_background.tex</code>	7	lat, degree, zip, son, second
Code	<code>s3_reduced_form.tex</code>	19	son, lon, lat, second, city, degree
Code	<code>s4_model.tex</code>	13	lon, lat, son, name, zip, house, city
Code	<code>s5_estimates.tex</code>	9	lon, son, lat, second
Code	<code>s6_counterfactual.tex</code>	16	lat, son, second, degree
Code	<code>s7_conclusion.tex</code>	2	lat, son
Code	<code>sa_add_figs.tex</code>	2	lon, lat
Code	<code>sb_firm_pricing.tex</code>	5	lon, son, lat
Code	<code>sc_robust.tex</code>	6	lon, second, zip, lat
Code	<code>sim_c0_sum_stat.R</code>	108	son, lat, lname, name, loc, zip
Code	<code>sim_c1_mh_regression.R</code>	111	lat, son, name, lname, lon
Code	<code>sim_c2_mh_viol.R</code>	25	lat, son, lname, name, loc, zip
Code	<code>sim_c3_demand_elast.R</code>	27	city, son, lat, lname, name, second, coord
Code	<code>sim_c4_data_figures.R</code>	24	lat, name, lname, son

File Type	File	Variables/References	PII Categories
Code	sim_c5_selection_figures.R	62	son, lat, lname, name, loc, zip, lon
Code	sim_c6_estimate.R	35	lat, son, name, block, loc, lname
Code	sim_c7_model_fit.R	62	lat, lon, block, loc, name, son, second
Code	sim_c8_ctf_run.R	59	lat, name, son, block, loc, second, lon
Code	sim_generate_data_list.R	158	son, lat, name, lname, lon, block, loc, zip
Code	sim_generate_rf_data.R	104	lat, son, name, lname, loc, zip, city, sex
Code	sim_helpers.R	58	lat, son, name, lname

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-04-02T12:26:14.530

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (<code>C:\</code> etc)
114	2	21	23	1

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/__MACOSX/Yizhou Jin - jinvas26_replication_mar_29_2026/code/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c

Filename	Size (MB)	Checksum (MD5)
/__MACOSX/Yizhou Jin - jinvas26_replication_mar_29_2026/data/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/Yizhou Jin - jinvas26_replication_mar_29_2026/data/estimates/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/Yizhou Jin - jinvas26_replication_mar_29_2026/paper/._.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

There are three provided R run files, including a master “run all” file.

cloc		github.com/AIDanial/cloc v 2.06 T=0.32 s (340.5 files/s, 679829.6 lines/s)		
Language	files	blank	comment	code
CSV	36	0	0	186570
JSON	13	0	0	13110
R	41	1990	1856	12134
TeX	17	244	15	1309
Markdown	1	97	0	279
Text	1	4	0	17
---	---	---	---	---
SUM:	109	2335	1871	213419

☒ The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

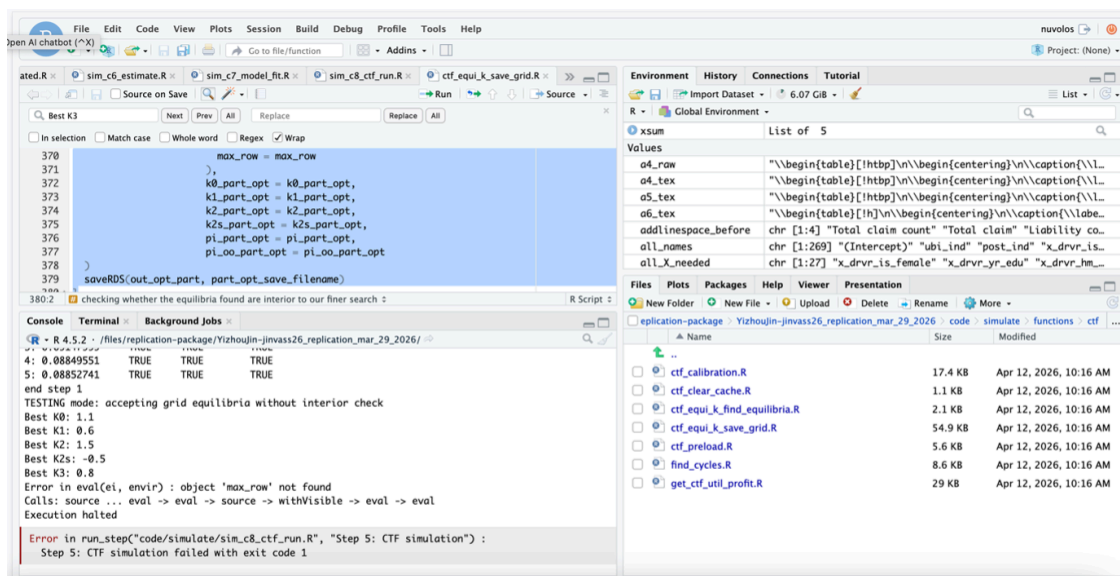
- RStudio 2026.01 + R 4.5.2 + Quarto + JDK 21 242.8 MB
- Mac Laptop M2 MacBook Pro, MacOS Monterey, 8 GB of memory, (M1 Processor)

8 Replication

- Downloaded code and data from dropbox.
- Committed code to git repo.
- Ran code as per README, but Step 5 (CTF simulation) in `run_simulated.R` failed when executing `sim_c8_ctf_run.R`. The error message was:

```
Error in eval(ei, envir): object 'max_row' not found
```

- The error originates from a call chain:
 - `run_simulated.R` calls `simulate/sim_c8_ctf_run.R` on line 164
 - then `simulate/sim_c8_ctf_run.R` calls to `simulate/functions/ctf/ctf_equi_k_save_grid.R` on line 295, where the variable `max_row` is used but not defined.
- This was resolved by adding `max_row <- max_row_coarse` to line 445 in `ctf_equi_k_save_grid.R`, after `max_row_fine = 2`
- The issue was not documented in the replication package, and no log files were generated for simulation steps, so the error only appeared in console output:



- Ran again and code ran to completion.

9 Findings

The replication reproduced the overall structure of the paper and the qualitative results in the tables and figures, including consistency between data and model outputs across panels.

Differences in the levels of the plotted values in several of the figures presenting simulated data were observed (figures in the Appendices), but these were expected (simulated data is designed to generate qualitatively similar but numerically different results):

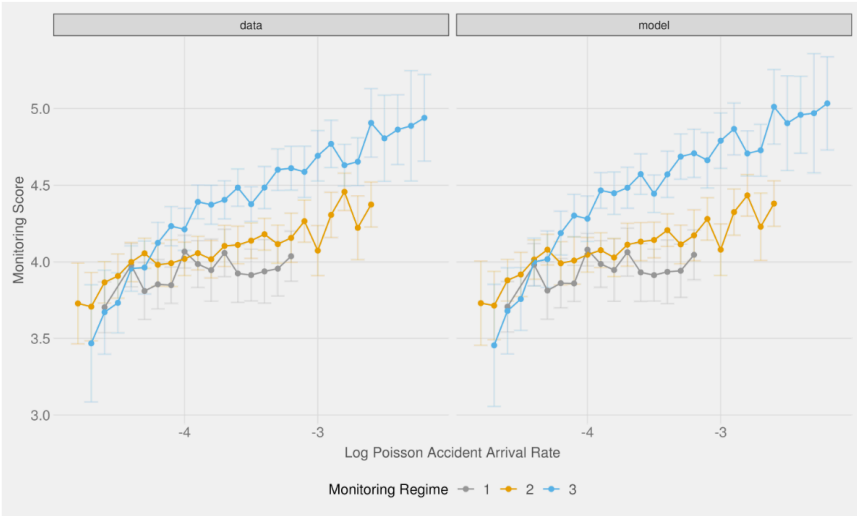


Figure B.3: Monitoring Score vs. Accident Risk by Monitoring Regimes

Notes: The X-axis represents the estimated Poisson risk arrival rate while the Y-axis plots monitoring score, data or predicted by our model, by monitoring regimes. Figure 6(b) corresponds to monitoring regime 3 in the right panel.

Replicated Figure B.3



Figure B.3: Monitoring Score vs. Accident Risk by Monitoring Regimes

Notes: The X-axis represents the estimated Poisson risk arrival rate while the Y-axis plots monitoring score, data or predicted by our model, by monitoring regimes. Figure 6(b) corresponds to monitoring regime 3 in the right panel.

Original Figure B.3

9.1 Missing Requirements

- ☐ Software Requirements
 - ☐ Stata
 - ☐ Version
 - Packages go here
 - ☐ Matlab

- ☐ Version
- ☒ R
 - `doParallel`

Had to manually install 'doParallel' package during Step 5 (CTF simulation) in `run_simulated.R`:

Error in `library(doParallel)`: there is no package called 'doParallel'

```

run_all.R x run_simulated.R x
Source on Save Run Source
Step 4 Next Prev All Replace Replace All
In selection Match case Whole word Regex Wrap
156 cat("\n--- Step 4: Estimation skipped (RUN_ESTIMATION = FALSE) ---\n")
157 }
158
159 #####
160 ## Step 5: CTF simulation (requires CmdStan + estimation output)
161 #####
162
163 if (RUN_CTF) {
164   run_step("code/simulate/sim_c8_ctf_run.R", "Step 5: CTF simulation")
165 } else {
166   cat("\n--- Step 5: CTF skipped (RUN_CTF = FALSE) ---\n")
167 }
168
169 #####
170 ## Step 6: Generate exhibits (shared code with real-data mode)
171 #####
167:2 (Untitled) R Script

Console Terminal Background Jobs
R - R 4.5.2 - /files/replication-package/Yizhoujin-jinvas26_replication_mar_29_2026/
code/simulate Next Prev Case sensitive

Step 4: Running brand and cost calibration...
Error in library(doParallel) : there is no package called 'doParallel'
Calls: source -> withVisible -> eval -> eval -> library
Execution halted

Error in run_step("code/simulate/sim_c8_ctf_run.R", "Step 5: CTF simulation") :
  Step 5: CTF simulation failed with exit code 1
Show Traceback
Rerun with Debug

> install.packages("doParallel")
  
```

9.2 Data Preparation Code

- Program `run_all.R` ran without error, output expected data
- Program `run_simulated.R` ran with the errors documented above (package and code errors). Once fixed, output expected data

9.3 Tables and Figures

All figures and tables in the main text of the paper were successfully generated.

The following exhibits were not generated in Section II, `run_simulated.R`:

- Tables A.9–A.15
- Tables C.2, C.4, C.5

The log printed in the RStudio console was as follows:

```
> source("code/c3_get_fitctf_exhibits.R")

c3_get_fitctf_exhibits.R: Generating model fit + CTF exhibits...
[GEN] tab_4.tex
[GEN] tab_5.tex
[SKIP] appendix/tab_c2.tex (no CSV)
[GEN] fig_6a.png
[GEN] fig_6b.png
[GEN] appendix/fig_b3.png
[GEN] appendix/fig_b4.png
[GEN] tab_7.tex
[SKIP] appendix/tab_a9.tex (no CSV)
[SKIP] appendix/tab_a10.tex (no CSV)
[SKIP] appendix/tab_a11.tex (no CSV)
[SKIP] appendix/tab_a12.tex (no CSV)
[SKIP] appendix/tab_a13.tex (no CSV)
[SKIP] appendix/tab_a14.tex (no CSV)
[SKIP] appendix/tab_a15.tex (no CSV)
[SKIP] appendix/tab_c4.tex (no CSV)
[SKIP] appendix/tab_c5.tex (no CSV)

c3_get_fitctf_exhibits.R completed: 7 generated, 10 skipped
```

Data Preparation Program	Dataset created	Reproduced?
run_simulated.R	Simulated datasets (JSON profiles, model inputs)	Yes
code/simulate/sim_c8_ctf_run.R	Counterfactual simulation outputs	Yes (conditional on RUN_CTF)
code/c1_get_rf_exhibits.R	Reduced-form datasets for figures/tables	Yes
code/c2_get_param_tables.R	Model parameter tables (estimation outputs)	Partial (some outputs missing)
code/c3_get_fitctf_exhibits.R	Model fit and counterfactual exhibit datasets	Partial (some CSVs not generated)

Figure/Table	Program	Line Number	Reproduced?
Table 1	c1_get_rf_exhibits.R	—	Yes
Table 2	c1_get_rf_exhibits.R	—	Yes
Table 3	c1_get_rf_exhibits.R	—	Yes
Table 4	c3_get_fitctf_exhibits.R	—	Yes
Table 5	c3_get_fitctf_exhibits.R	—	Yes
Table 6	c2_get_param_tables.R	—	Partial (simulated differences)
Table 7	c3_get_fitctf_exhibits.R	—	Partial (simulated differences)
Figure 2a	c1_get_rf_exhibits.R	—	Yes

Figure/Table	Program	Line Number	Reproduced?
Figure 2b	c1_get_rf_exhibits.R	—	Yes
Figure 3	c1_get_rf_exhibits.R	—	Yes
Figure 4	c1_get_rf_exhibits.R	—	Yes
Figure 5	c1_get_rf_exhibits.R	—	Yes
Figure 6a	c3_get_fitctf_exhibits.R	—	Partial (simulated differences)
Figure 6b	c3_get_fitctf_exhibits.R	—	Partial (simulated differences)
Fig 1a–d	static	—	No
Table A.1	c1_get_rf_exhibits.R	—	Yes
Table A.3–A.6	c2_get_param_tables.R	—	Partial (simulated differences)
Table A.7–A.8	c2_get_param_tables.R	—	No
Table A.9–A.15	c3_get_fitctf_exhibits.R	—	No (CSV missing)
Fig A.1–A.2	static	—	No
Fig A.3–A.6	c1_get_rf_exhibits.R	—	Yes
Fig B.1–B.4	mixed (c1/c3)	—	Partial (simulated differences)
Table C.1	c1_get_rf_exhibits.R	—	Yes
Table C.2–C.3	c3_get_fitctf_exhibits.R	—	Partial (simulated differences)
Table C.4–C.5	c3_get_fitctf_exhibits.R	—	No (CSV missing)
Fig C.1–C.4	c1_get_rf_exhibits.R	—	Yes
In-text numbers	Program	Line Number	Reproduced?
—	—	—	

9.4 In-Text Numbers

- ☐ In-text numbers not verified.
- ☒ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code

10 Classification

- ☐ full reproduction
- ☒ full reproduction with minor issues
- ☐ partial reproduction (see above)

- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☒ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
- ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements